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Stochastic Modelling in Finance and Insurance

Problem Set 3

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Instructions

Please submit your answers as a **text-based file** (PDF or Word) containing all required deliverables, such as numerical results, figures, and brief discussions. Add your properly annotated and executable **R code**. You are encouraged to use code snippets from the course and **Generative AI** for tutoring, coding, and writing. However, please remember that over-reliance on AI may compromise your learning effect. Finally, all work must be your own; copying from others is strictly prohibited.

Exercise 1: Nelson-Siegel-Svensson Model

a) Use the interest rate data in the file *YieldCurve.csv* to estimate the Nelson-Siegel-Svensson model. **Report:** Your parameter estimates $(\beta_0, \beta_1, \beta_2, \beta_3, \lambda_1, \lambda_2)$.

b) Based on your estimated model parameters from a), program an R function that returns the spot interest rate $y_0(T)$ for any maturity T.
Report: A plot of your spot yield curve.

c) Use your spot yield curve function from b) to program an R function for arbitrage-free forward rates $f_0(T_1, T_2)$ with an effective date T_1 and a termination date T_2 ($T_1 < T_2$). **Report:** A plot of the forward curve for effective date $T_1 = 1$ (over the whole spectrum of termination dates T_2).

d) Based on your estimated model parameters from a), program an R function that returns the instantaneous forward rate $f_0(T, T + dt)$ for any effective date T.
Report: A plot of your instantaneous forward curve.

Exercise 2: Hull-White Model

Consider a Hull-White model for the short rate

$$\mathrm{d}\tilde{r}_t = (\theta_t - \alpha r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}\widetilde{W}_t$$

with mean reversion level

$$\theta_t = \frac{\partial f_0(t, t+dt)}{\partial t} + \alpha f_0(t, t+\mathrm{d}t).$$

Assume a mean reversion speed $\alpha = 0.20$ and short-rate volatility $\sigma = 0.01$. All rates and volatilities are expressed in decimal form, consistent with the Chapter 4 R code. The instantaneous forward curve is the same as in **Exercise 1)**.

a) Simulate 1000 short rate paths over 10 years with a step size of 250 trading days per year. **Report:** A plot of your paths (include the instantaneous forward curve).

b) **Report:** A histogram of the short rate in one year, in five years and in ten years. Overlay the probability density function (pdf) of a normal distribution with a matching mean and standard deviation. What do you see?

c) **Report:** The *expected* price of a zero-coupon bond with a notional of 100 and a maturity of one year in three years from now.

d) Consider a *caplet*, which is a call option on a future interest rate $\tilde{y}_{t_k}(t_{k+1})$ with cap rate (strike rate) y_X and payoff date t_{k+1}. Its payoff is:

$$c_{t_{k+1}} = N\,\Delta t_k \max\left(\tilde{y}_{t_k}(t_{k+1}) - \mathrm{y_X}, 0\right),$$

with

$\Delta t_k = t_{k+1} - t_k$

Interest rate effective date t_k:	2 years
Option payoff date t_{k+1}:	3 years
Notional N:	1'000'000
Cap rate (strike rate) y_X:	1.5%.

Report: The price of the caplet under the traditional risk-neutral measure $\mathbb{Q}_*$

[**Hint:** Do not confuse $\mathbb{Q}_*$ with the forward measure $\mathbb{Q}_{(t_k, t_{k+1})}$.]

Exercise 3: GenAI in Financial Engineering

Use GenAI (e.g., ChatGPT, Gemini) to create an innovative derivative product on interest rate risk. Pick your own assumptions for the Hull-White model and price the product, using R and your knowledge from the course.

Report: Your prompts, an intuitive description of the product, its formal payoff function, and its price.